



Contoso Engineering (illustrative mock)

AI-SDLC roadmap G2

Plan, build and review

ASSESSMENT DATE	2026-09-27
FRAMEWORK VERSION	2.0.1
RESPONDENTS	14
GENERATED ON	2026-09-28

G2: Plan, build and review

ID	DIMENSION	SCORE	LEVEL	TARGET	GAP	PRIORITY	HORIZON
D3	Plan, Specify and Design	2.09	L2 Adopting	3.00	0.91	P2 Medium	Semester
D4	Code and Context Engineering	2.72	L3 Scaling	3.00	0.28	P3 Low	Backlog / monitor
D5	Review, Quality and Testing	2.10	L2 Adopting	3.00	0.90	P2 Medium	Semester

D3: Plan, Specify and Design

in agentic coding, "people make most of the planning decisions (what to do) and Claude makes most of the execution decisions (how to do it)" [24]; the quality of the task definition drives the quality of agent output [8], [27].

Start here

D3-Q1 AI in backlog refinement (2.00, L2)

L3 target: Most teams use AI to draft or improve work items; acceptance criteria are mandatory before work starts.

Then L4: Work-item quality (clarity, testability) is measured and linked to rework and cycle time. [16][17]

D3-Q2 Specification before implementation (2.00, L2)

L3 target: Plans or specs are stored in the repository or linked to the issue, and are reviewed by a human before agent implementation.

Then L4: Specs are the contract for automated verification (tests, checks) and are kept in sync with the code. [23][24][27][58]

D3-Q5 User-centric focus (2.07, L2)

L3 target: Work items reference the user problem and success measure; feedback is reviewed before prioritization.

Then L4: User-outcome metrics are part of the definition of done for AI-assisted delivery. [1][2][3]

Recommended strategies

- S5 GitHub Copilot Acceleration (P2 Medium, Semester)
- S3 App Modernization (P1 High, Next quarter)

All questions in this dimension

ID	QUESTION	SCORE	N	NA	EVIDENCE
D3-Q1	AI in backlog refinement	2.00	13	1	2
D3-Q2	Specification before implementation	2.00	12	2	6
D3-Q3	Task scoping for agents	2.15	13	1	3
D3-Q4	Architecture and design decisions	2.15	13	1	4
D3-Q5	User-centric focus	2.07	14	0	4
D3-Q6	AI-assisted modernization	2.14	14	0	7

D4: Code and Context Engineering

GitHub measures adoption depth as a progression from "Code first" to "Agent first" to "Multi-agent" [6]; Anthropic describes context as "a finite resource with diminishing marginal returns" [25].

Start here

D4-Q4 Repository instructions (2.50, L3) **unverified**

L3 target: Most active repositories have structured instructions (build, test, conventions, constraints) with an owner; changes go through PR review; files are updated when the codebase changes rather than committed once.

Then L4: Instructions are generated from a shared baseline, checked for staleness, and their effect on agent merge rate and code quality is measured, since instruction files alone do not guarantee better results. [2][8][25][27][46][47][55]

D4-Q1 Depth of AI use across surfaces (2.62, L3) **unverified**

L3 target: Many engineers regularly use at least one GitHub agent surface ("Agent first"), confirmed by usage metrics.

Then L4: Multi-agent use is normal ("Multi-agent"), with cohort distribution tracked monthly. [6]

D4-Q2 Agent mode for multi-file work (2.64, L3)

L3 target: Agent mode is the default for multi-file refactors and features in most teams; changes are reviewed in the diff before commit.

Then L4: Teams share agent-mode patterns that work and track where it fails; tool permissions are tuned per repository. [6][27]

Recommended strategies

- S5 GitHub Copilot Acceleration (P2 Medium, Semester)
- S6 Agentic Activation (P2 Medium, Semester)

All questions in this dimension

ID	QUESTION	SCORE	N	NA	EVIDENCE
D4-Q1	Depth of AI use across surfaces unverified	2.62	13	1	5
D4-Q2	Agent mode for multi-file work	2.64	14	0	7
D4-Q3	Coding agent delegation unverified	2.67	12	2	4
D4-Q4	Repository instructions unverified	2.50	12	2	5
D4-Q5	Reusable prompts, agents and skills unverified	2.79	14	0	4
D4-Q6	MCP server governance unverified	2.79	14	0	5
D4-Q7	AI access to internal knowledge unverified	3.08	12	2	3
D4-Q8	Model selection and routing	2.69	13	1	7

D5: Review, Quality and Testing

DORA links AI-driven change volume to instability unless strong control systems exist [3]; GitHub requires human review before agent PRs merge [7]; 46% of developers distrust AI output accuracy [37].

Start here

D5-Q6 Verification culture and calibrated trust (1.77, L2)

L3 target: Review guidelines explain what to check in AI output; trust in AI output is part of the developer survey.

Then L4: Trust and accuracy are compared with real defect data, and guidance is updated where they diverge. [3][23][35][37][48][52]

D5-Q3 Same quality gates for AI and human code (2.00, L2)

L3 target: Required checks are enforced by rulesets on the protected branches of most repositories, with no bypass for agent identities.

Then L4: Gates are policy as code, applied across all repositories (>90%) and reviewed after incidents. [3][7][31][50][55]

D5-Q1 AI-assisted code review (2.07, L2)

L3 target: AI review runs automatically on most PRs; teams track useful vs dismissed suggestions.

Then L4: Review rules are tuned per repository from suggestion outcomes; review time and escaped defects are tracked; PRs where only AI reviewed AI-authored code are visible and governed. [6][12][14][54][55]

Recommended strategies

- S5 GitHub Copilot Acceleration (P2 Medium, Semester)
- S7 Security & Governance (P2 Medium, Semester)

All questions in this dimension

ID	QUESTION	SCORE	N	NA	EVIDENCE
D5-Q1	AI-assisted code review	2.07	14	0	6
D5-Q2	Human-in-the-loop for agent changes	2.07	14	0	6
D5-Q3	Same quality gates for AI and human code	2.00	13	1	2
D5-Q4	Small batches	2.17	12	2	7
D5-Q5	AI-assisted testing	2.31	13	1	2
D5-Q6	Verification culture and calibrated trust	1.77	13	1	8
D5-Q7	Code health of AI-generated code	2.29	14	0	2

References

- [1] DORA. DORA AI Capabilities Model. Google Cloud, 2025. <https://dora.dev/ai/capabilities-model/report/>
- [2] DORA. DORA AI Capabilities Model: Survey Questions. Google Cloud, 2025. <https://dora.dev/ai/capabilities-model/questions/>
- [3] Google Cloud. Announcing the 2025 DORA Report: State of AI-Assisted Software Development. 2025. <https://cloud.google.com/blog/products/ai-machine-learning/announcing-the-2025-dora-report>
- [6] GitHub Docs. GitHub Copilot usage metrics. <https://docs.github.com/en/copilot/concepts/billing-and-usage/copilot-usage-metrics/copilot-metrics>
- [7] GitHub Docs. Risks and mitigations for GitHub Copilot cloud agent. <https://docs.github.com/en/enterprise-cloud@latest/copilot/concepts/security-governance-and-network-settings/risks-and-mitigations>
- [8] GitHub Docs. Best practices for using GitHub Copilot to work on tasks. <https://docs.github.com/en/enterprise-cloud@latest/copilot/tutorials/cloud-agent/get-the-best-results>
- [12] GitHub Docs. About GitHub Copilot code review. <https://docs.github.com/en/copilot/concepts/agents/code-review>
- [14] GitHub Docs. Application card: GitHub Copilot Agents. <https://docs.github.com/en/copilot/responsible-use/agents>
- [16] Microsoft Azure Blog. Agentic DevOps: Evolving software development with GitHub Copilot and Microsoft Azure. 2025. <https://azure.microsoft.com/en-us/blog/agentic-devops-evolving-software-development-with-github-copilot-and-microsoft-azure/>
- [17] Microsoft for Developers. Agentic DevOps in action: Reimagining every phase of the developer lifecycle. 2025. <https://developer.microsoft.com/blog/reimagining-every-phase-of-the-developer-lifecycle/>
- [23] Anthropic. Anthropic Economic Index: AI's impact on software development. 2025-04-28. <https://www.anthropic.com/research/impact-software-development>
- [24] Anthropic. How Claude Code is used in practice. 2026-06-16. <https://www.anthropic.com/research/claude-code-expertise>
- [25] Anthropic. Effective context engineering for AI agents. 2025-09-29. <https://www.anthropic.com/engineering/effective-context-engineering-for-ai-agents>
- [27] Anthropic. Best practices for Claude Code. Claude Code Docs. <https://code.claude.com/docs/en/best-practices>
- [31] Gartner. Gartner Says the Market for Enterprise AI Coding Agents Is Entering a New Phase of Expansion and Competitive Realignment. 2026-05-20. <https://www.gartner.com/en/newsroom/press-releases/2026-05-20-gartner-says-the-market-for-enterprise-ai-coding-agents-is-entering-a-new-phase-of-expansion-and-competitive-realignment>
- [35] METR. Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity. 2025-07-10. <https://metr.org/blog/2025-07-10-early-2025-ai-experienced-os-dev-study/> (paper: <https://arxiv.org/abs/2507.09089>)
- [37] Stack Overflow. 2025 Developer Survey: AI. <https://survey.stackoverflow.co/2025/ai>
- [46] Arabat, A., Sayagh, M. Toward Instructions-as-Code: Understanding the Impact of Instruction Files on Agentic Pull Requests. arXiv:2606.13449, 2026-06-11. <https://arxiv.org/abs/2606.13449>
- [47] Denisov-Blanch, Y., Agarwal, S., Azaletskiy, P., He, H., Schaeffer, R., Miranda, B., Vasilescu, B., Koyejo, S. A Few Pages of Markdown: Committed AI Configuration and Lower Quality Cost after Coding-Agent Adoption. arXiv:2608.25241, 2026-08-26. <https://arxiv.org/abs/2608.25241>
- [48] Shen, J. H., Tamkin, A. How AI Impacts Skill Formation. arXiv:2601.20245, 2026-01-28. <https://arxiv.org/abs/2601.20245> (Anthropic summary: <https://www.anthropic.com/research/AI-assistance-coding-skills>)
- [50] Takerngsaksiri, W., Duong, N., Barnett, S. Who Finishes the Job? A Study of Follow-Up Fixes and Commit Authorship on AI Coding Agent Pull Requests. arXiv:2609.26847, 2026-09-22. <https://arxiv.org/abs/2609.26847>
- [52] Sakib, A. H. M. N., Banik, D., Jadliwala, M. Trust but Verify? Uncovering the Security Debt of Autonomous Coding Agents. arXiv:2607.12428, 2026-07-14. <https://arxiv.org/abs/2607.12428>
- [54] Selvanayagam, N., Ghaleb, T. A. AI-to-AI Code Reviews of GitHub Pull Requests. arXiv:2608.21311, 2026-08-21. <https://arxiv.org/abs/2608.21311>

[55] Stolze, M., Strässle, M. When Review Alone No Longer Scales: Layered Supervision in AI-Assisted Software Engineering. arXiv:2608.26316, 2026-08-26. <https://arxiv.org/abs/2608.26316>

[58] Farrag, S. E. The Productivity-Reliability Paradox: Specification-Driven Governance for AI-Augmented Software Development. arXiv:2605.01160, 2026-05-01. <https://arxiv.org/abs/2605.01160>